

On This Night

*A Population-Scale Smartphone Sky Observatory — and What It
Can Honestly Do*

Aaron Garcia

aaron@garcia.ltd

Version 2.0 | PUBLIC

8 August 2026

Summary

Once a year, at one agreed instant, millions of people point their phones at the sky and take a photograph. That is the whole idea. What can be done with the result is the subject of this paper, and the honest answer is narrower than the idea first suggested — but not nothing, and worth stating precisely because an earlier version of this concept oversold it.

This is a concept paper. No app has been built, no event has been held, and no organisation has been contacted. It proposes a design and argues, from established physics, what such a dataset could and could not yield.

1. What It Is Not

The original framing called this a “synthetic aperture telescope”. It is not one, and cannot be. Optical synthetic-aperture imaging combines light while preserving phase to a fraction of a wavelength — of order ten nanometres at visible wavelengths. Consumer phones offer metre-scale GPS position, tens-of-milliseconds time synchronisation, and sit under an atmosphere whose coherence time is milliseconds. That is a mismatch of six to eight orders of magnitude, and it is fundamental physics, not an engineering gap no algorithm can close. This paper drops the synthetic-aperture claim entirely, because keeping it would have made everything else untrustworthy.

2. What Stacking Actually Buys

What survives is ordinary image stacking, and it is genuinely useful. Combine N photographs of the same patch of sky and the random noise averages down while the real signal stays: the signal-to-noise ratio improves as the square root of N . Ninety thousand phones in one stadium, all pointed at the same zenith, stack to roughly a three-hundred-fold noise improvement; a million images of one field would give a thousand-fold. Faint structure invisible in any single grainy phone frame emerges from the average.

There is a distinction the earlier draft blurred, and it is the crux of the whole proposal, so it is worth stating flatly. Stacking for a deep image requires the same patch of sky. People standing together — a stadium, a city — share very nearly the same zenith and can be stacked into one deep frame. People spread across the globe do not: each sees a different sky, so their images cannot be co-added into a single deep picture. The global dataset is therefore not one deep image. It is something else — simultaneous coverage of the entire night hemisphere from millions of viewpoints — and its value comes from that simultaneity and that spread, not from deep-stacking. Conflating the two is the single easiest way to overstate this project, and this version does not.

Resolution, separately, is capped by the atmosphere: even perfect stacking cannot beat the one-to-two-arcsecond blur of astronomical seeing, and phone optics sit above that already. Stacking cleans noise; it does not sharpen.

3. Where It Is Not First: Unistellar

Honesty about prior art matters more here than anywhere. A coordinated citizen network that observes the same targets simultaneously and stacks the results into composite images already exists and already publishes science: Unistellar's network of tens of thousands of smart telescopes, run with the SETI Institute and NASA, has produced composite super-images (its Comet ATLAS campaign) and peer-reviewed results including a Nature paper tied to the DART impact. Coordinated citizen stacking is therefore not new, and any honest version of this proposal must say so first.

What is different, and is the real and narrower contribution, is the instrument and the scale. Unistellar uses dedicated smart telescopes costing roughly two thousand pounds each, numbering in the tens of thousands. This proposal uses the phone already in a person's pocket, and aims at participation numbers two or three orders of magnitude larger. That changes what is possible — not deeper images than a real telescope, which a phone cannot match, but a density and geographic spread of simultaneous viewpoints that a fleet of expensive instruments cannot reach. The contribution is coverage and distribution, not aperture.

4. What Population-Scale Distribution Uniquely Enables

Three applications follow from having millions of simultaneous, geographically spread observers — and depend on the spread, not on deep images, so they sidestep the Unistellar comparison entirely.

Application	Why distribution is the enabling ingredient	Confidence
Space-debris and satellite parallax	Observers hundreds of km apart see the same low-orbit object against different star backgrounds; the parallax triangulates its 3D position. Baseline length sets precision, so population spread is the asset	Geometrically sound; precision unquantified [TARGET]
Atmospheric tomography	Millions of simultaneous sightlines through the upper atmosphere allow a volumetric reconstruction (airglow, gravity waves); smartphone citizen geoscience has real precedent (Semeter et al., ionospheric TEC via phone GNSS)	Plausible, with funded precedent
Transient detection	A brief flash — a bright meteor, a satellite glint, a rare optical transient — that one telescope would miss is caught somewhere among millions of simultaneous frames	Sound in principle; false-positive rejection is the open problem

These are the parts a phone swarm can do that a small number of good telescopes cannot, and they are where the honest novelty lives.

5. The Design

The event is deliberately simple, because citizen-science participation collapses when protocols get complicated. One night a year, one instant (with a few minutes' grace for latecomers), a short burst of raw photographs, automatic capture of location and

pointing from the phone's own sensors, and an unhurried upload over the following days on Wi-Fi while charging. Timing does not need to be exact: the science is designed around hundred-millisecond windows, not millisecond simultaneity, because that is what phone clocks and mobile networks actually deliver. Different devices are handled statistically, by building correction profiles per phone model from the aggregate rather than asking anyone to calibrate anything.

6. How It Could Fail, and the First Test

The reachable pilot is small: a single coordinated event with a few thousand committed participants — an astronomy-community night, not a hundred million strangers — with pre-committed criteria. Does a co-located subset stack to the predicted square-root-N noise improvement on a known star field? Can the distributed subset triangulate a known satellite's orbit to a stated accuracy? If the co-located stack does not reach its predicted SNR, or the parallax cannot recover a catalogued orbit, the technical premise is wrong and the project should stop there — cheaply. The hundred-million-participant figure in the original draft is not a plan; it is a ceiling that, the adoption literature makes clear, needs either a crisis catalyst or platform pre-installation, and this paper claims no route to it.

7. Invitation

The useful next step is small and testable, and the useful collaborator is someone who can run it: an astronomer or citizen-science group willing to hold one coordinated capture and check whether the stacking and the parallax deliver what the physics predicts. If they do not, that is worth knowing. Nothing here should be treated as demonstrated until that pilot has been run, and no partnership or funding described as a possibility should be read as one that exists — none has been sought.

Contact: aaron@garcia.ltd

Appendix A: Document Metadata

Document Title	On This Night
Subtitle	A Population-Scale Smartphone Sky Observatory — and What It Can Honestly Do
GUID	AG-2026-0808-6612
Version	2.0
Date	8 August 2026
Author	Aaron Garcia
Affiliation	Independent Researcher
Contact	aaron@garcia.ltd
Origin Context	Revised from a December 2025 draft; synthetic-aperture claim removed, repositioned against existing citizen-astronomy networks
Classification	PUBLIC

Status	RELEASED
--------	----------

Appendix B: Glossary

Term	Definition
Atmospheric seeing	Blurring of astronomical images by atmospheric turbulence, ~1–2 arcseconds at good sites
Parallax	Apparent shift of a nearby object against a distant background when viewed from different positions; enables triangulation
SNR	Signal-to-noise ratio; improves as the square root of the number of stacked images
Stacking	Combining many images of the same field so random noise averages out
Synthetic aperture	Combining signals with preserved phase to mimic a larger instrument — feasible in radio, not with consumer optical devices
TEC	Total Electron Content — an ionospheric measurement obtainable from dual-frequency GNSS, including modern smartphones
Tomography	Reconstructing a volume from many crossing sightlines, as in a CT scan

Appendix C: Methodology and Transparency Statement

Author Context and Expertise

This work is grounded in 30 years of multi-sector experience in Systems Engineering. The author is not an astronomer; the physical arguments here are deliberately elementary so that astronomers can check them.

Digital Methodology and Accessibility

In the spirit of transparency, I utilised a suite of Generative AI tools — specifically Claude, Google Gemini, and ChatGPT — to assist in the production of this paper. These tools were employed as “force multipliers” for data synthesis and editorial accessibility. As a writer with dyslexia, I leverage these models as assistive technology to refine grammar and streamline sentence structure.

Verification and Integrity

An earlier draft framed this project as a “synthetic aperture telescope”, which is physically impossible for consumer devices; that claim has been removed and the reason explained in the text. Nothing in this paper has been built, run, or tested, and no person or organisation has been contacted — the partnership and funding routes discussed are candidate future steps, not arrangements that exist. The prior work it stands against (notably the Unistellar/SETI/NASA citizen network) is real and is credited as prior art, not as a collaborator.

Appendix D: References

Citation	Reference
Unistellar / SETI / NASA	Unistellar Citizen Science network (with the SETI Institute and NASA): coordinated simultaneous observation and composite stacking; peer-reviewed results including a Nature paper on the DART impact. seti.org/projects/unistellar-network .
Semeter et al.	Smartphone dual-frequency GNSS for ionospheric Total Electron Content (NASA/NSF citizen-science precedent), Boston University. See NSF EAGER award 2428531 and related Space Weather publications, 2024–2025.
Google Pixel astrophotography	Multi-frame stacking for night-sky imaging in Google Pixel cameras; underlying methods published in ACM Transactions on Graphics.
GLOBE at Night	Citizen-science sky-brightness programme; a template for low-friction participation design.
Astrometry.net / Lang et al.	Blind astrometric calibration of arbitrary astronomical images — relevant to registering heterogeneous citizen photographs.